Syed Aabis Akhtar

Aerospace Engineering

Propulsion hardware, flight systems, CAD, autonomy and engineering software.

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Website

aab.is

GitHub

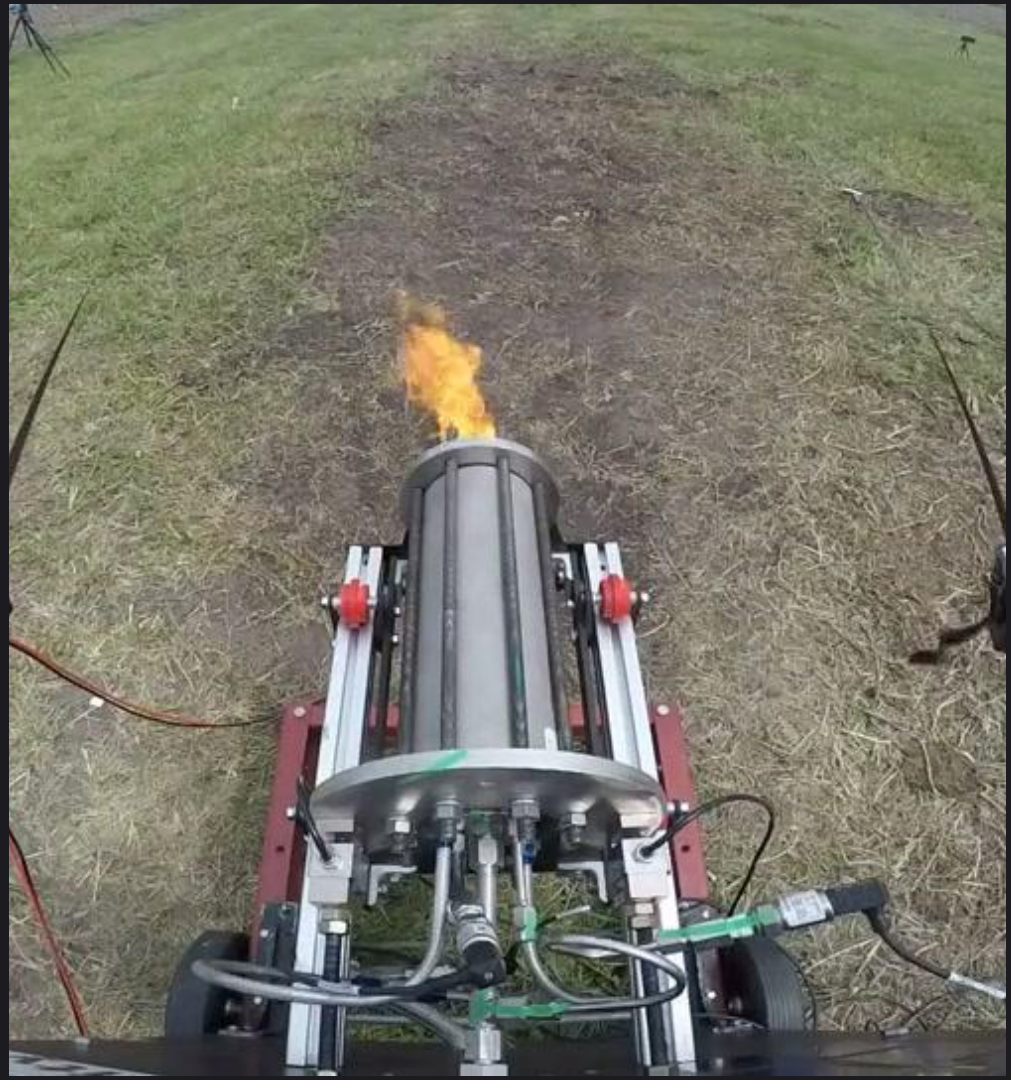
github.com/GitBolt

LinkedIn

linkedin.com/in/saabisa

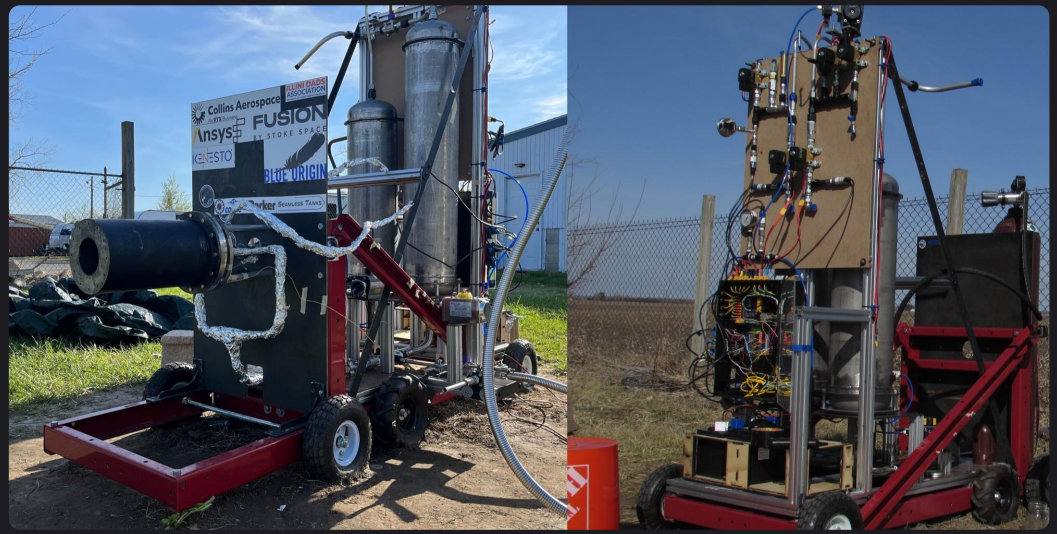
University of Illinois Urbana-Champaign

B.S. Aerospace Engineering · May 2029



Liquid Rocket Feed System

RAND-E test stand | Liquid Rocketry at Illinois



Student-built engine test stand (RAND-E)



Overture Mark 1 rocket engine

Fuel and oxidizer plumbing

Built the IPA/LOX feed from the team's piping and instrumentation diagram. Cut, bent and flared tubing. Installed valves, fittings and actuators on the RAND-E stand.

Test operations

Ran pressurization, leak checks and water flows that cleared the stand for hotfire. As Admin Director I interviewed test leads after tests and helped with key club decisions.

Liquid Rocket Engine Hotfire

Liquid Rocketry at Illinois | May 2026



Cold-flow test



Hot-fire sequence · May 3, 2026

Operations

Oxygen cleaning, cold-flow, hotfire

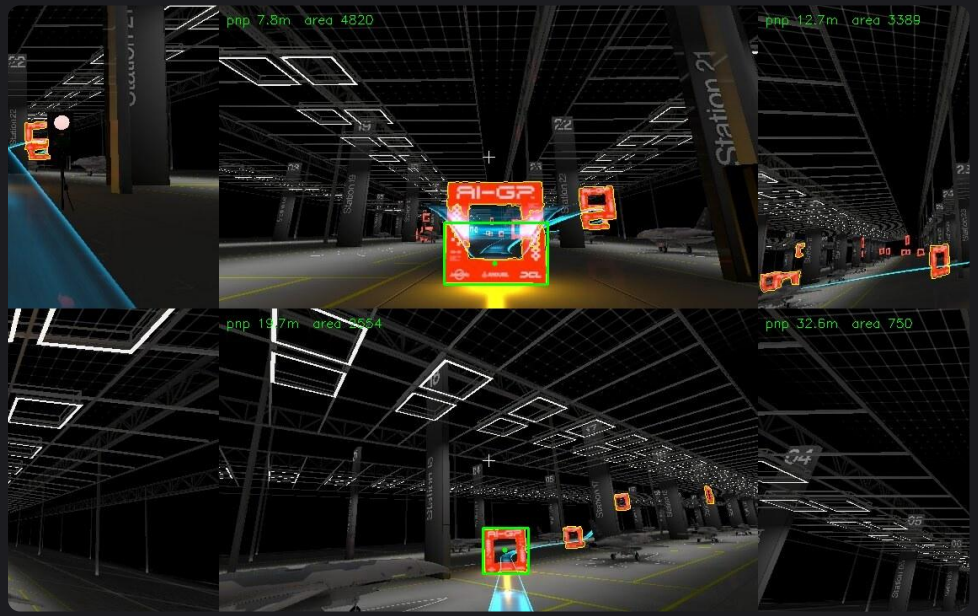
IPA / LOX

isopropyl alcohol / liquid oxygen

Helped with oxygen cleaning, leak testing and torqued fittings before cold-flow and hotfire. The engine ignited, but a controller fault cut the run early before the 10-second target. Second hot-fire aimed for October 2026.

Anduril Autonomous Drone Racing

Cleared Round 1 on Anduril's simulator | GPS-denied OpenCV/IMU



Gate racing view in Anduril simulator

Perception → control	
30 Hz	FPV camera + MAVLink telemetry
OpenCV / PnP	gate detection and pose
400 Hz	gate-relative state estimate
50–60 Hz	control loop with gated PPO policy

100%
easy-course completion

88%
medium-course completion

62%
hard-course completion

Around 7 m/s
Zero crashes

Carbon-fiber Quadcopter

Personal UAV | ArduPilot / MAVLink | hardware and flight testing



Workshop build



Assembled airframe showcase at a conference



Outdoor night flight

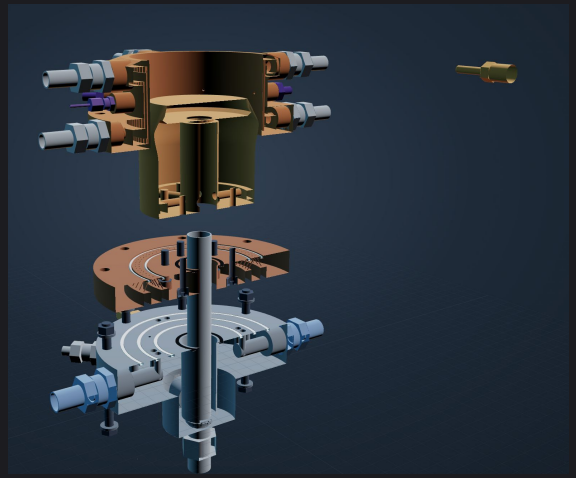
~2 kg

assembled total weight

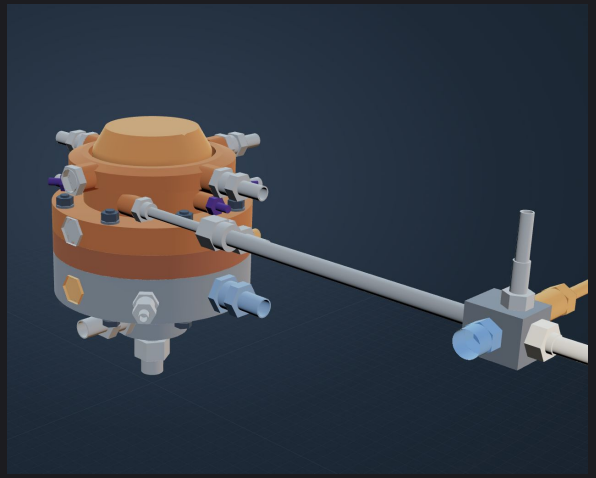
Built a ~2 kg carbon-fiber airframe and integrated motors, power, radio and flight-control hardware. Used ArduPilot and MAVLink to configure and control the aircraft. Bench-tested and flew outdoors.

Rotating Detonation Rocket Engine

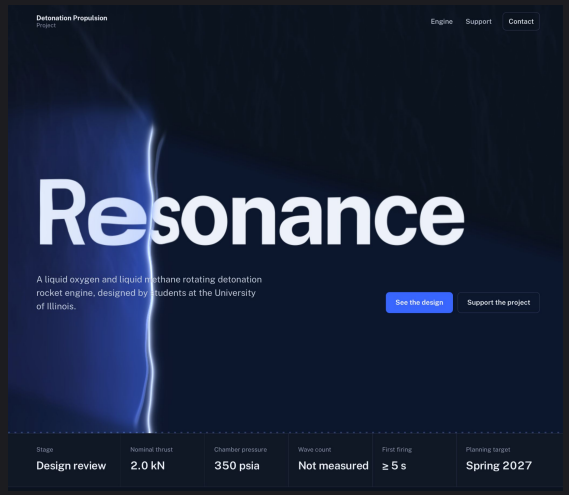
Founded student group for RDRE | Team Lead | LOX/LCH_4 space propulsion



Resonance Gen-1 engine internal CAD



Full Injector & Chamber Assembly



Public website at RDRELab.com

~2 kN
design thrust class

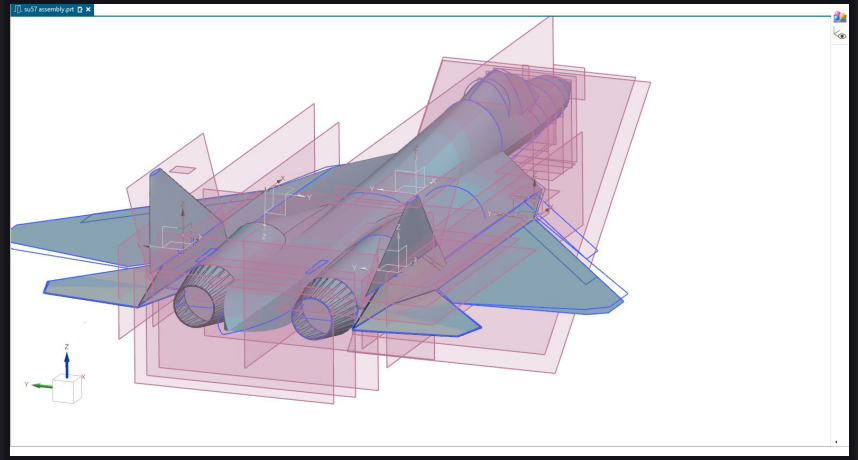
O/F 3.0
350 psia design point

Founded student group to work on RDRE, aiming to become the first student team to build one of the most efficient rocket engines for space vehicles.

Ran thermodynamic equilibrium calculations in NASA CEA to define sizing parameters & operating points. Went from idea to team, finalizing design, and securing manufacturing sponsorships in 3 weeks.

SU-57 CAD model

Siemens NX



CAD model in NX with datum planes and sketches



Rendered CAD assembly with decals and materials

01 DATUMS

Datum planes and full-span / length checks

02 SKETCH

Traced fuselage and planform profiles

03 SURFACE

Through-curves and extruded geometry

04 ASSEMBLY

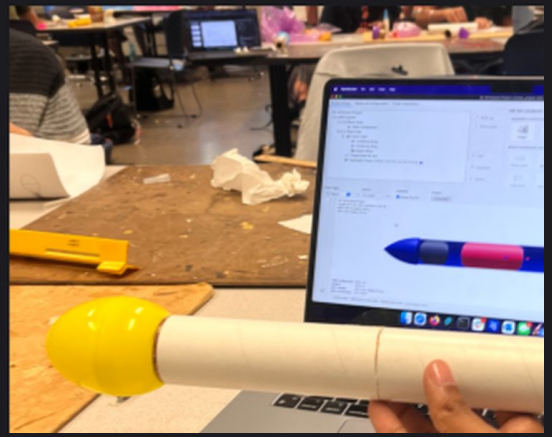
Mirrored features, wing and rudder .prt parts

Model Rocket: Design and Launch

AE 100 course project | OpenRocket simulation and field launch



October 2025 launch footage



OpenRocket build · model and simulation



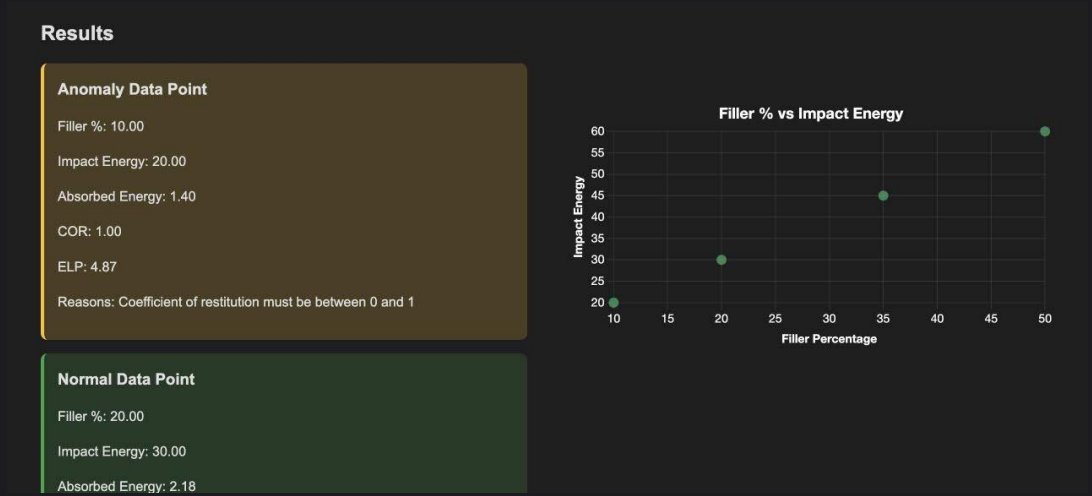
Predicted altitude vs time

381 mm tube · 50.8 mm OD · balsa fins · 56 g egg · D12-3 · 305 mm chute

Designed the rocket in OpenRocket and launched it in the field. The first sim showed large pitch, yaw and angle of attack. After a stability revision, the model predicted 106.4 m apogee and 46.5 m/s peak speed at 179 g launch mass.

Syntactic Foam Impact Analyzer

NMCAD Lab research internship | Indian Institute of Science



Built the kNN analysis pipeline and a review interface

kNN model for foam impact data

Key capabilities & workflow:

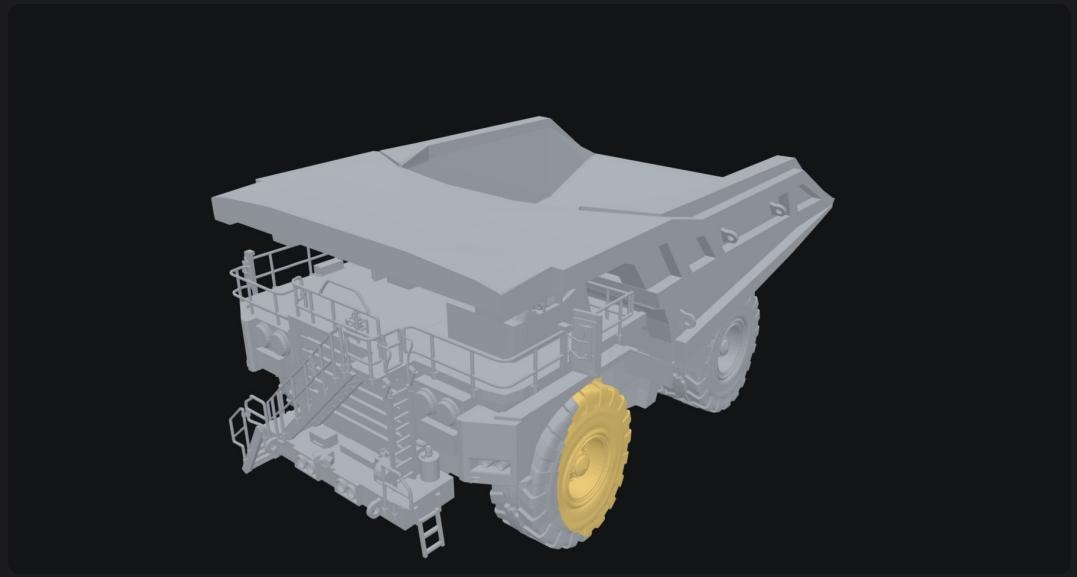
- Groups samples by filler fraction percentage
- Evaluates absorbed energy & restitution (COR)
- Flags nonphysical predictions (COR > 1)

Sample checks & anomaly detection

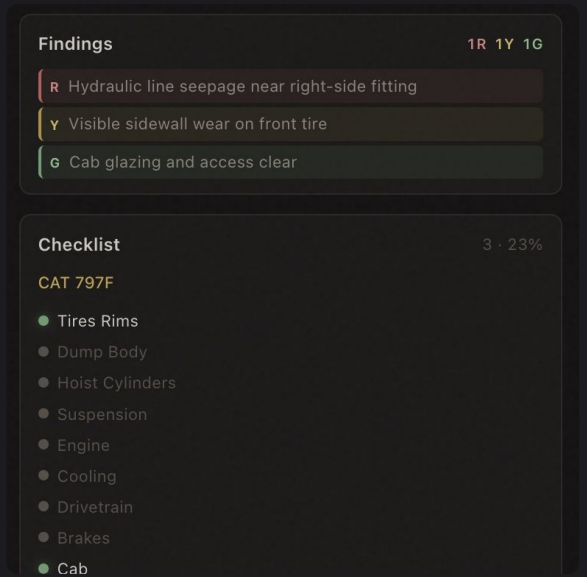
At IISc NMCAD Lab under Prof. Dineshkumar Harursampath, I built a Python kNN model for syntactic-foam energy absorption. It groups samples by filler fraction and flags nonphysical predictions. Sample checks can flag cases where predicted restitution exceeds 1. I also helped redesign the lab site.

CatWatch: Inspection Software

HackIllinois 2026 | Caterpillar heavy-vehicle inspection



CAT 797F model in 3D defect viewer



Sample findings inspection checklist

Python SDK

WebSocket camera stream

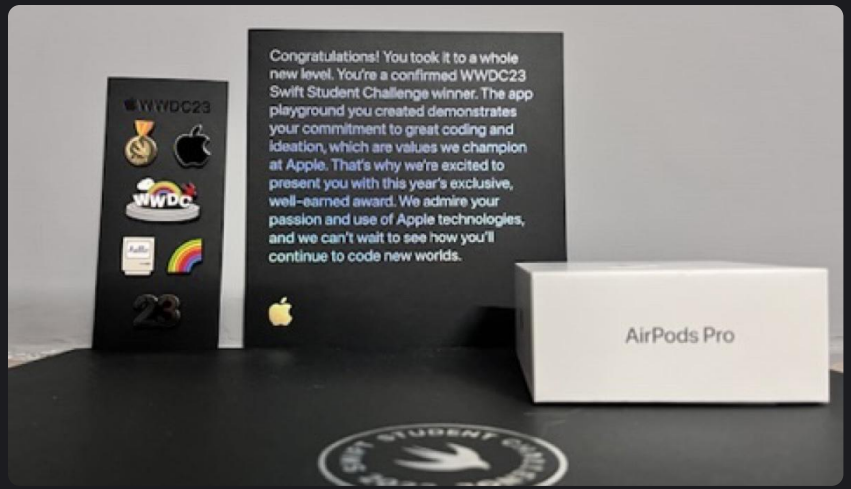
Modal CV

Cloud inference and API-key portal

Built a Python SDK that streams camera frames over WebSocket into Modal cloud CV with an API-key portal. Detections map to a 3D defect viewer for Caterpillar at HackIllinois 2026.

Selected Software Projects

150,000+ downloads | Swift Student Challenge



Winning prizes sent by Apple from Menlo Park



Python Solana SDK with downloads statistics

AsymmED

Built an iOS app that explains asymmetric encryption. Won Apple's 2023 Swift Student Challenge after finishing in four days while learning Swift.

Solathon

Published a typed Python SDK for Solana RPC and transactions with sync and async clients. Crossed 150,000+ downloads and received a $3,000 grant for development.